Lecture 2: Supply and Demand (Chapter 2)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, ch. 2 (*Supply and Demand*). · Lecturer: **Luigi Butera.**

Theme of the lecture (Butera): build **tractable models of demand and supply**, then use them to find the **market equilibrium** — the price at which "*no one would benefit from changing their own behaviour given the choices of others.*" The whole of micro is understanding that equilibrium: what sets prices and quantities, and how they move.

> ★ **The heart of the chapter:** a market is one picture — a **downward-sloping demand curve** and an **upward-sloping supply curve** that cross at **exactly one point, the equilibrium (p*, q*)**, where **quantity demanded = quantity supplied**. Three skills: ① write demand & supply as **functions**; ② set them equal to **solve for p* and q***; ③ **shift** a curve and read off the new equilibrium. *Price is set by BOTH blades of the scissors (Marshall) — never demand or supply alone.*

The red thread. For *every* question ask the one diagnostic that trips people up: **is this a *movement ALONG* a curve (only the good's *own price* changed) or a *SHIFT of* the whole curve (something *else* changed — income, other prices, costs, tastes, tech)?** Own price → slide along. Anything else → the curve moves. Get that right and the rest is algebra.

Key concepts / "modes" to use

- **Demand function** `Q_D = D(p, pₛ, p_c, Y, τ)` → its 2-D **demand curve**; **inverse demand** (p as a function of q).
- **Supply function** `Q_S = S(p, p_o)` → **supply curve**; **inverse supply**.
- **Movement along vs shift** of a curve (the core distinction); **shifters** of each curve.
- **Substitutes / complements**; **normal / inferior** goods; **Giffen** good (the rare exception).
- **Market equilibrium** `Q_D = Q_S` → solve for **p*, q***; **excess demand (shortage) / excess supply (surplus)**.
- **Comparative statics** (shifts) & how the **slope** of the other curve sizes the effect.
- **Price ceiling / floor**; **perfect competition** assumptions; **Marshall's scissors**.

Section-by-section main points

2.1 Demand — the demand function

Core: demand answers "*how much do buyers want at each price?*" The **law of demand:** all else equal, **lower price → higher quantity demanded** (the curve slopes **down**).

- *In plain terms — why it slopes down:* two reasons. ① **Scarcity** — you don't have infinite money, so a lower price lets you afford more. ② **Diminishing marginal utility** — the *first* slice of pizza is worth a lot to you; the 30th almost nothing, so you'll only buy more if it's cheaper.
- **The demand function** describes quantity demanded `Q_D` as a function of price **and everything else**:

DEMAND FUNCTION

$Q\_D = D(\ p\ ,\ p_s\ ,\ p\_c\ ,\ Y\ ,\ \tau\ )$ — p = own price · p_s = price of a **substitute** · p_c = price of a **complement** · Y = income · τ = taxes

- **How the 2-D curve appears — "flush" the other variables into a constant.** A demand *curve* plots `Q_D` against **own price only**, so you **fix** everything else at set numbers; they collapse into the intercept. The book's worked reduction — start from a specific form and plug in temperature T=20°, substitute price p_r=2, income I=5:

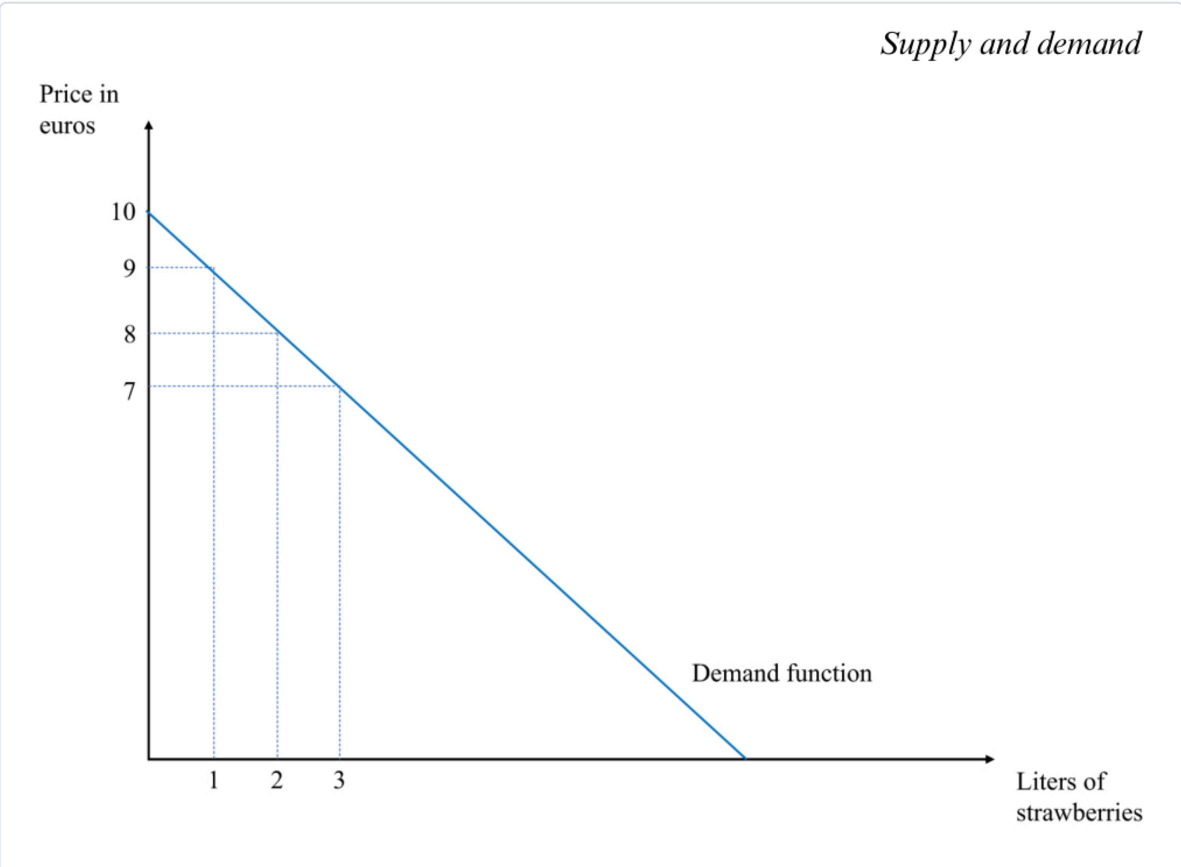
COLLAPSING TO A LINEAR DEMAND CURVE

$$Q\_D = 1 - p + 0.25 \cdot T + 0.75 \cdot p_r + 0.5 \cdot I = 1 - p + \mathbf{5 + 1.5 + 2.5 = 10 - p}$$
(intercept $a = 10$ bundles all the non-price factors)

LINEAR DEMAND & ITS INVERSE (FOR GRAPHING)

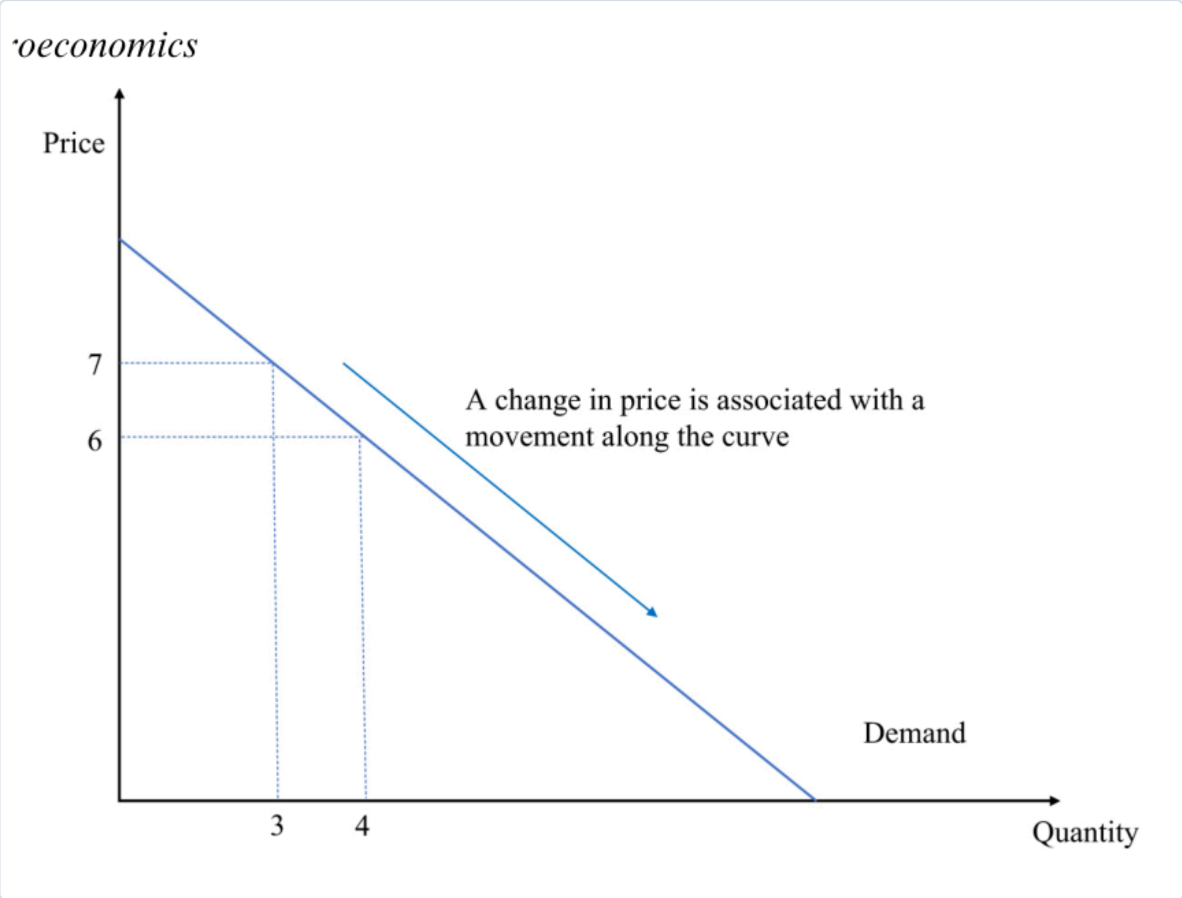
$Q\_D = a - b \cdot p \rightarrow$ e.g. $\mathbf{Q\_D = 10 - p}$ $(a = 10,\ b = 1)$. Solve for $p \Longrightarrow$ **inverse demand: $\mathbf{p = 10 - Q\_D}$** · *(deck's version: $Q\_D = 20 - 2p \Longrightarrow p = 10 - ½ \cdot Q\_D$)*

Check: p = 7 → Q_D = 3; p = 6 → Q_D = 4. **a** = the intercept (shift it and the whole curve moves); **b** = how many units demand falls per +1 in price.



How to read it (the demand curve). By convention **price is on the vertical axis, quantity on the horizontal** (blame Alfred Marshall). Each point answers *both* questions: "at price p, how much is bought?" *and* "for quantity q, what's the most a buyer will pay?" Here the line hits **p = 10 when q = 0** (nobody buys above €10) and slides down to more litres as price falls.

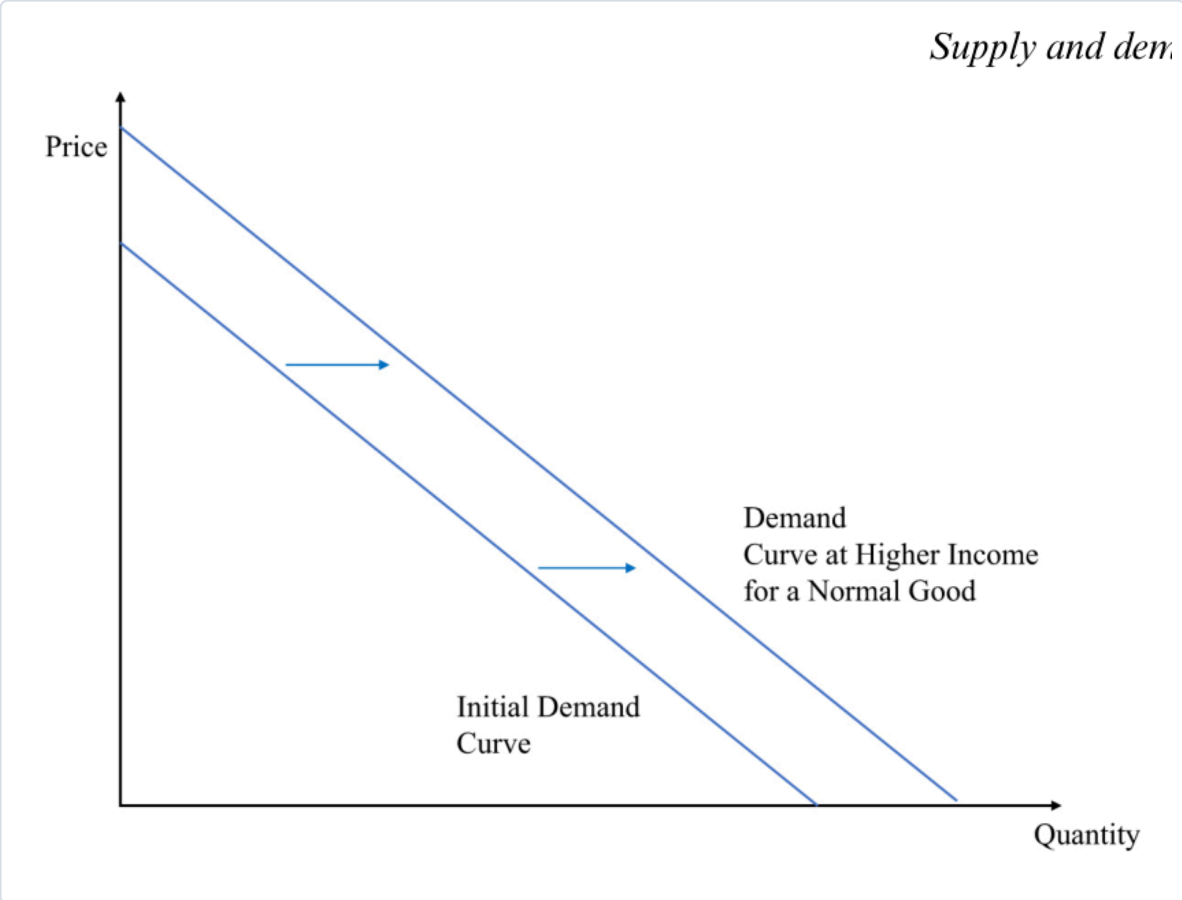
- ★ **Movement ALONG vs SHIFT — the make-or-break distinction:**
  - **Own price changes → MOVE ALONG** the existing curve (you slide to a new point).
  - **Anything else changes → the whole curve SHIFTS** (a *new* curve).

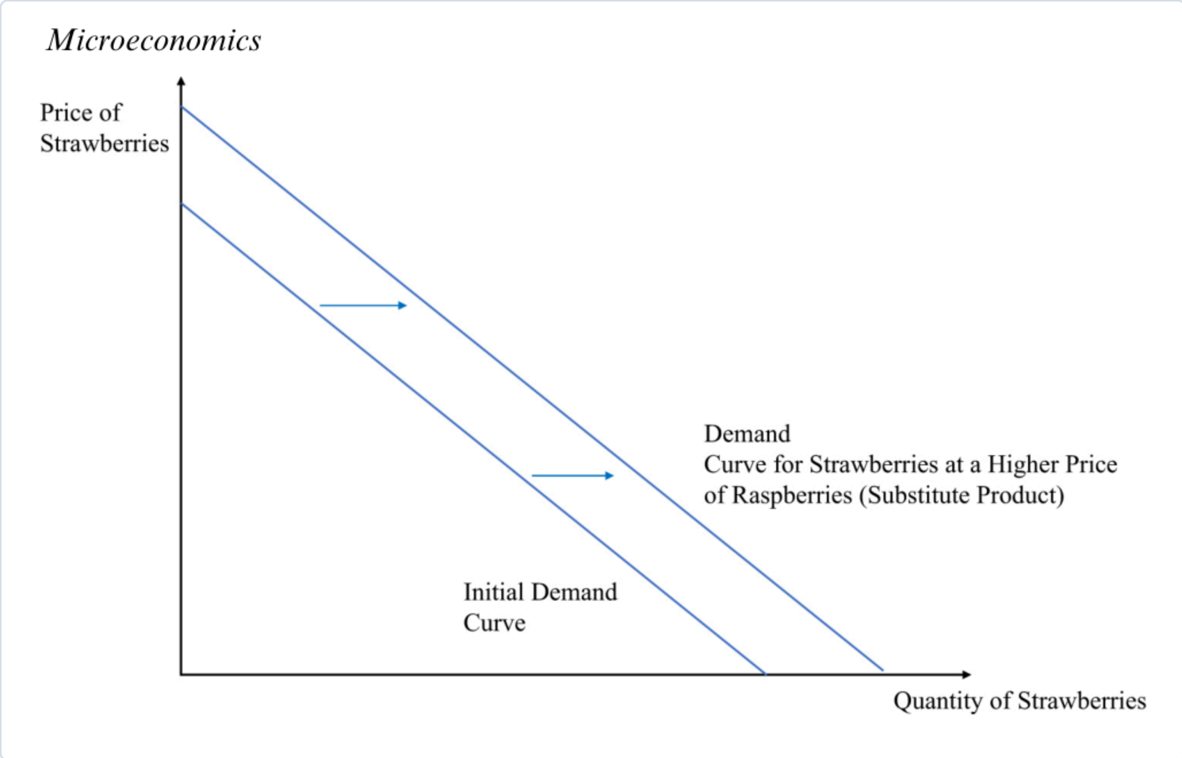


How to read it (movement along). Only the **own price** moved (7→6), so you **stay on the same curve** and just slide to a new point (q 3→4). No new curve is drawn. Contrast the shifts below, where the curve itself jumps.

- **Demand shifters (each moves the *whole* curve = changes the intercept a):**

Shifter	Direction	Why
Income ↑ (normal good)	shift **out** (right)	richer → buy more at every price
Income ↑ (inferior good, e.g. instant noodles)	shift **in** (left)	richer → buy *less* of it
Price of a substitute ↑ (raspberries ↑ → strawberry demand)	shift **out**	switch toward the now-relatively-cheaper good
Price of a complement ↑ (cereal ↑ → milk demand)	shift **in**	the pair is consumed together
Tastes / expectations (heatwave → ice-cream; COVID → toilet paper)	out or in	common-sense direction





How to read them (shifts). The whole line moves **right (out)** = more demanded at *every* price. Fig 2.4: higher **income** (normal good). Fig 2.6: a **substitute** (raspberries) got dearer, so buyers swing to strawberries. Note you are *not* sliding along the old curve — a brand-new curve sits to the right.

- **Substitutes vs complements (definitions): substitutes** — demand for good 1 **rises** when good 2's price rises (raspberries ↔ strawberries). **Complements** — demand for good 1 **falls** when good 2's price rises (milk ↔ cereal).
- **From individual to aggregate demand (deck).** Market demand = **horizontal sum** of every buyer's demand — *add the quantities at each price*:

AGGREGATE (MARKET) DEMAND

$$Q(p) = q_1 + q_2 + \dots = D_1(p) + D_2(p) + \dots \textit{ (add quantities across buyers at the same price)}$$

Deck example: Mike's inverse demand $p = 100 - 2q\_M$, Linda's $p = 100 - ½q\_L$ → invert each to $q(p)$, then add: at each price, total $q = q\_M + q\_L$. *(Watch for a **kink** where one buyer drops out at high prices.)*

- **The T-shirt trap (don't confuse products):** seeing pricey Gucci shirts outsell cheap generics is **not** an upward-sloping demand curve — those are **different goods**. A demand curve compares a good **to itself** (Gucci cutting *its own* price sells *more*). Real upward-sloping demand = the very rare **Giffen good** (Irish-famine potatoes; poor Chinese rice households — Jensen & Miller 2008).

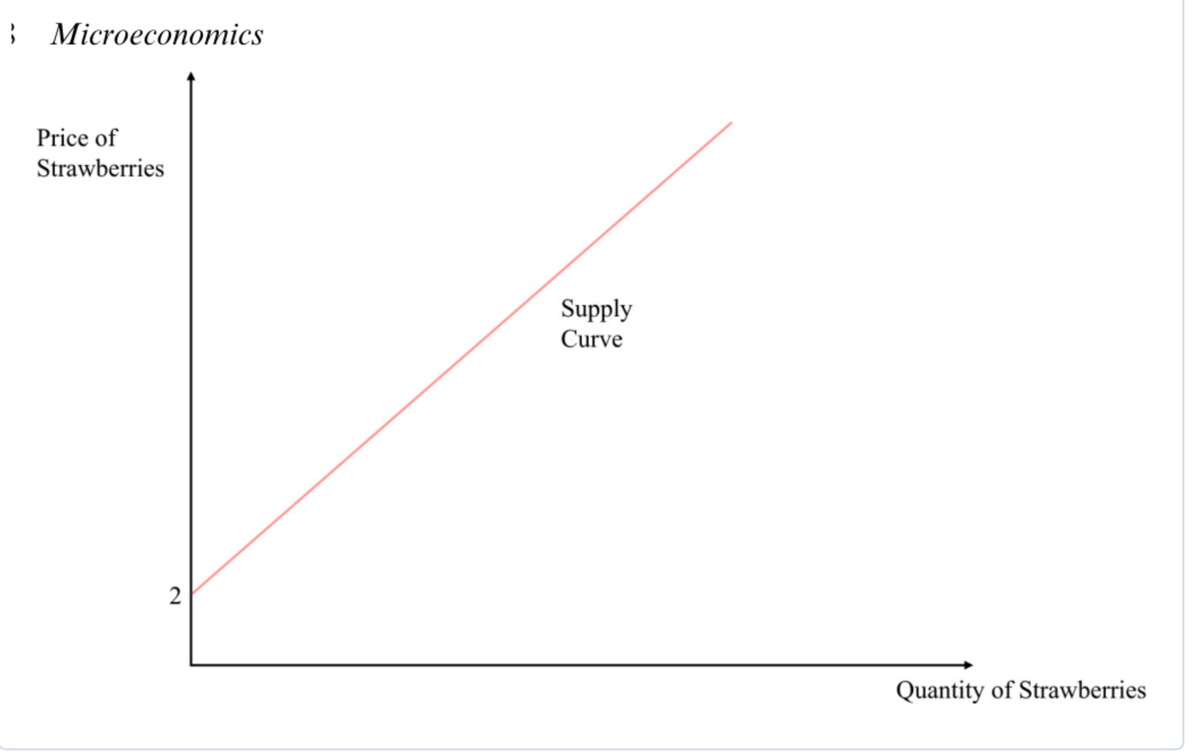
2.2 Supply — the supply function

Core: supply answers "*how much do firms bring to market at each price?*" Firms are **price-takers** (they choose *quantity*, not the price). The curve usually slopes **up** — higher price → more supplied.

- *In plain terms — why upward:* a higher price (i) makes it worth producing more even as costs rise, and (ii) draws in higher-cost producers who couldn't profit at the low price.
- **The supply function** (quantity supplied depends on price and input/other factors):

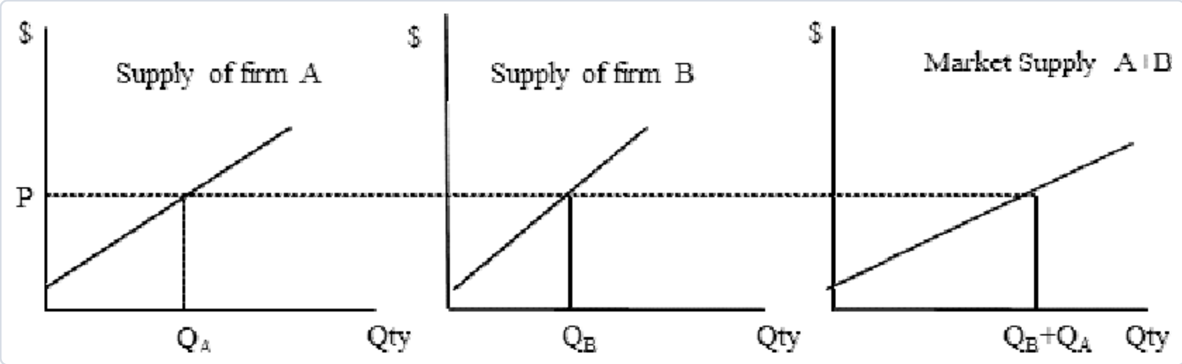
> SUPPLY FUNCTION & ITS INVERSE
>
> $Q\_S = S(\ p\ ,\ p\_o\ )$ — $p\_o =$ price of inputs. Book's linear example: $\mathbf{Q\_S = -2 + p} \implies$ inverse supply $\mathbf{p = 2 + Q\_S}$

- **Why the "−2" (a negative intercept that isn't weird):** read it in **inverse** form — **price must clear €2 before *any* is supplied** (at p = 2, Q_S = 0). *Check:* p = 3 → Q_S = 1; p = 4 → Q_S = 2.



How to read it (the supply curve). Start at the **€2 price-intercept** (below it, zero supply) and climb: each extra euro of price brings one more litre to market. Like demand, an **own-price** change = **move along**; anything else = **shift**.

- **Supply shifters (move the whole curve): input/cost ↑** (wages, transport) → shift **in** (need a higher price for the same quantity); **cheaper inputs / new tech / more sellers / market opens to foreign suppliers** → shift **out**; **disruptions** (strikes, extreme weather, mine closures) → shift **in**.
- **From individual to aggregate supply (deck):** market supply = **horizontal sum** of each firm's supply — add quantities across firms at each price.



How to read it (aggregate supply). At any price **P**, read each firm's quantity off its own curve (Q_A, Q_B) and **add them across** — the market curve is the sum, so it's **flatter** (more responsive) than any single firm's.

★ 2.3 Market equilibrium — solving for p* and q*

Core: equilibrium = the **one price where quantity demanded = quantity supplied** (`Q_D = Q_S`), the **market-clearing** price — no shortage, no glut, nobody wants to move. **This is the calculation to master.**

- **The recipe (memorise these three steps):**

STEP 1 — SET DEMAND = SUPPLY

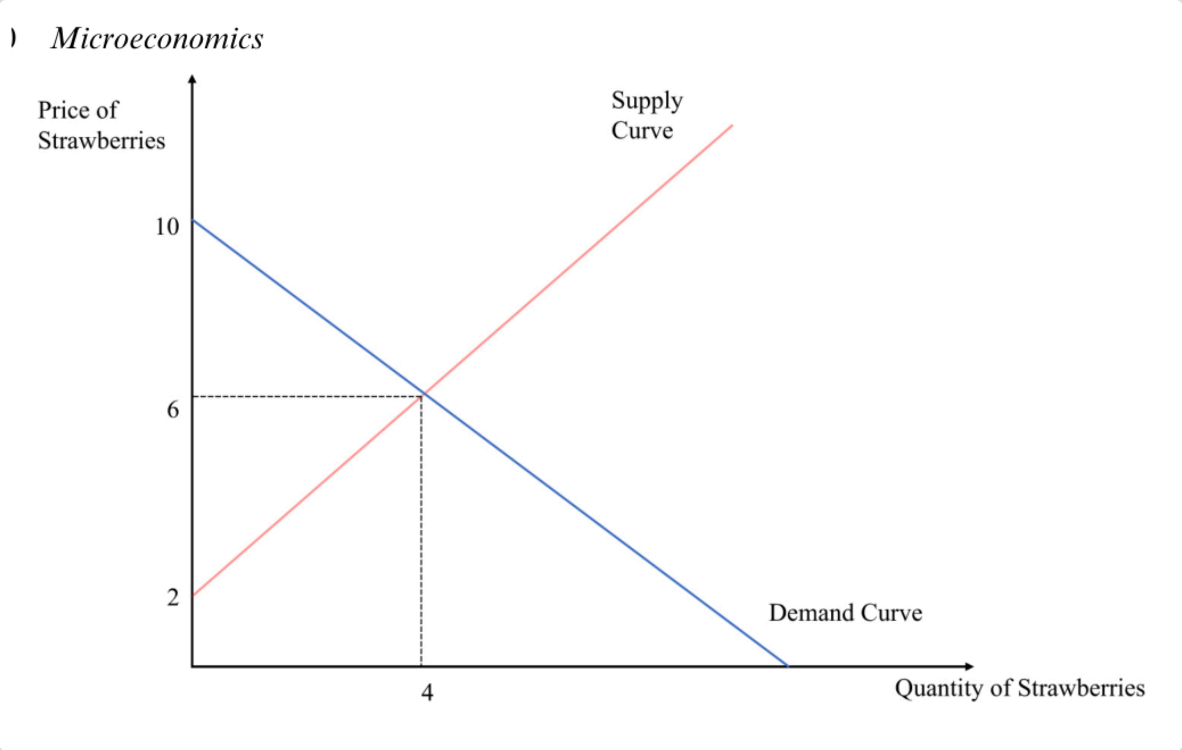
$$Q\_D = Q\_S \implies 10 - p = -2 + p$$

STEP 2 — SOLVE FOR THE EQUILIBRIUM PRICE P*

$$10 + 2 = p + p \implies 12 = 2p \implies \mathbf{p^* = 6}$$

STEP 3 — PUT P* BACK INTO EITHER CURVE FOR Q*

$$Q^* = 10 - 6 = \mathbf{4} \text{ (check: } Q\_S = -2 + 6 = 4 \checkmark\text{)} \implies \mathbf{equilibrium\ (p^*, q^*) = (6, 4)}$$



How to read it (equilibrium). The **crossing point** is the only place the two plans agree — buyers want exactly what sellers offer (4 litres at €6). Everywhere else, one side is frustrated and price gets pushed toward the cross.

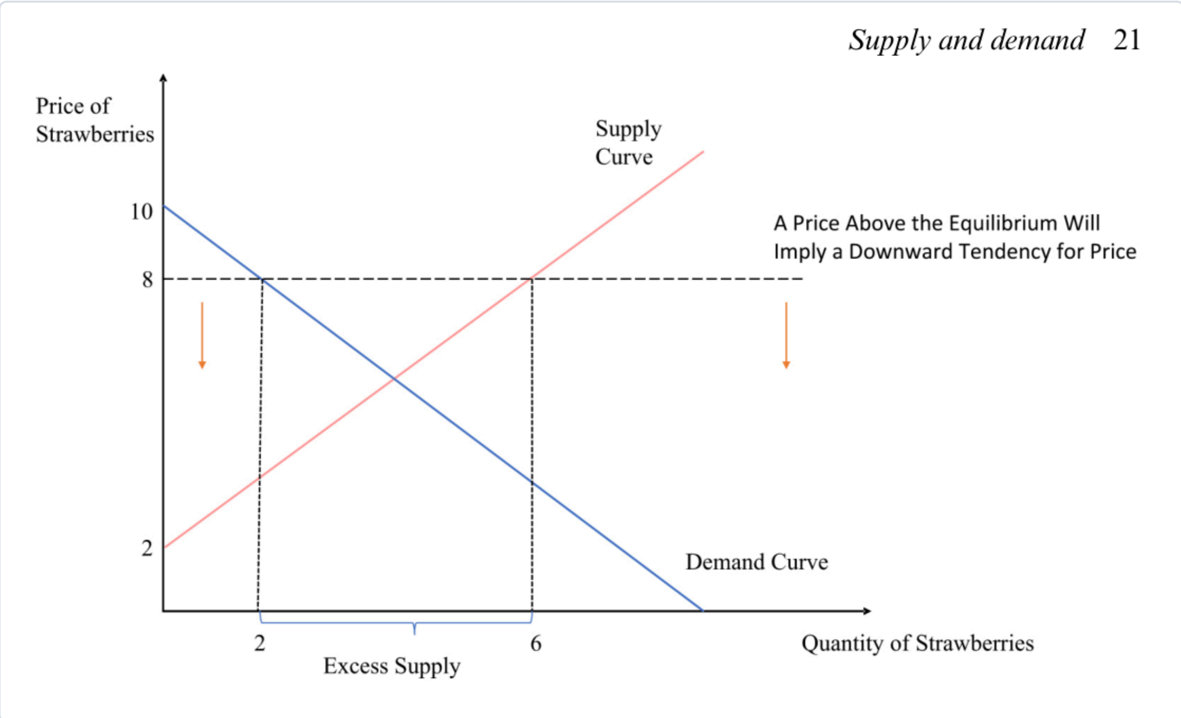
- **Off-equilibrium: excess demand & excess supply** (this is also *exactly* how you handle price controls below):

EXCESS DEMAND (SHORTAGE) — PRICE BELOW P*

$$\text{at p} = 4\text{: Q\_D} = 10–4 = \mathbf{6}\text{, Q\_S} = -2+4 = \mathbf{2} \implies \text{shortage} = \text{Q\_D} - \text{Q\_S} = \mathbf{4} \to \text{upward pressure on price}$$

EXCESS SUPPLY (SURPLUS) — PRICE ABOVE P*

$$\text{at p} = 8\text{: Q\_D} = 10–8 = \mathbf{2}\text{, Q\_S} = -2+8 = \mathbf{6} \implies \text{surplus} = \text{Q\_S} - \text{Q\_D} = \mathbf{4} \to \text{downward pressure on price}$$



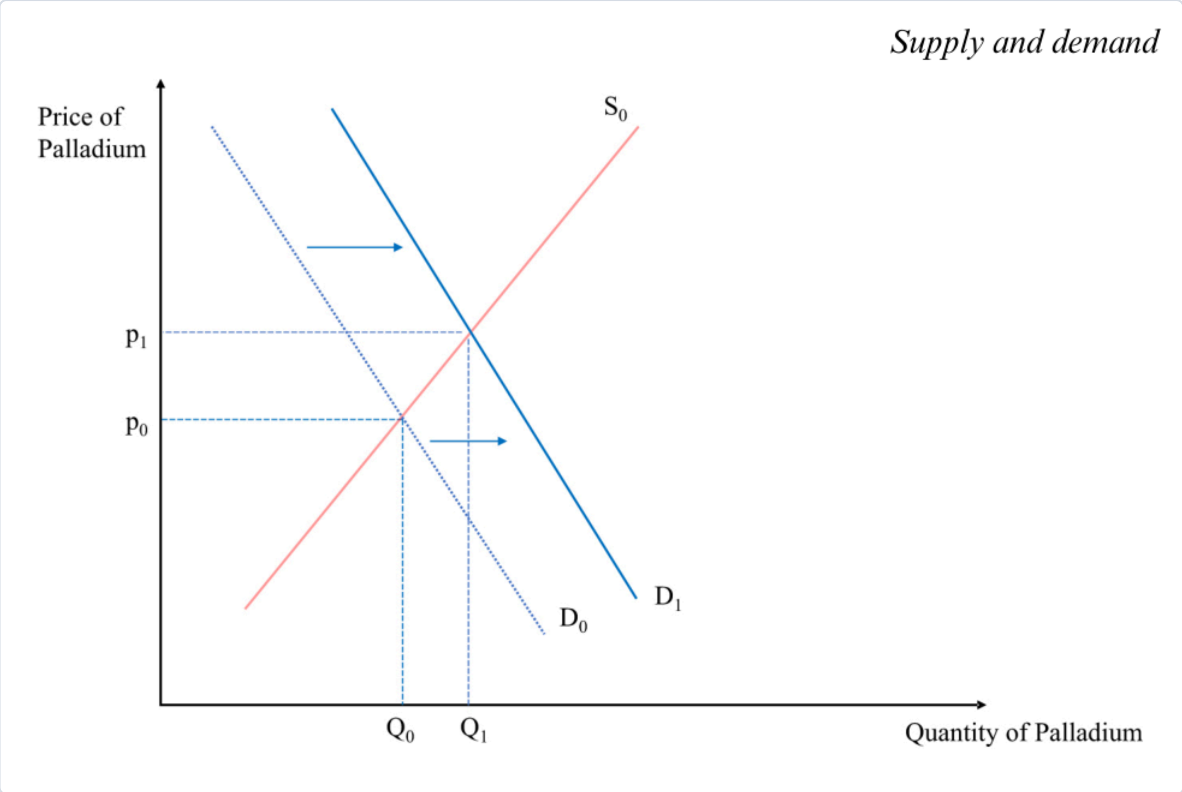
How to read them. Pick the off-equilibrium price, drop a vertical line, and read **both** curves: the **horizontal gap** between them is the shortage (demand side wider) or surplus (supply side wider). The gap is what pushes price back toward p*.

- **Marshall's scissors:** asking whether *demand or supply* sets the price is like asking *which blade of a scissors cuts* — **both do.** This dissolves the **diamond–water paradox**: water is cheap not because it's little wanted but because it's **abundant in supply**; diamonds are dear because they're **scarce**.
- *(Who actually "sets" the price if everyone's a price-taker? The theoretical **Walrasian auctioneer** — and **Vernon Smith's 1962 experiments** showed real people reach the equilibrium price fast.)*

2.4 Shifts in the equilibrium — comparative statics

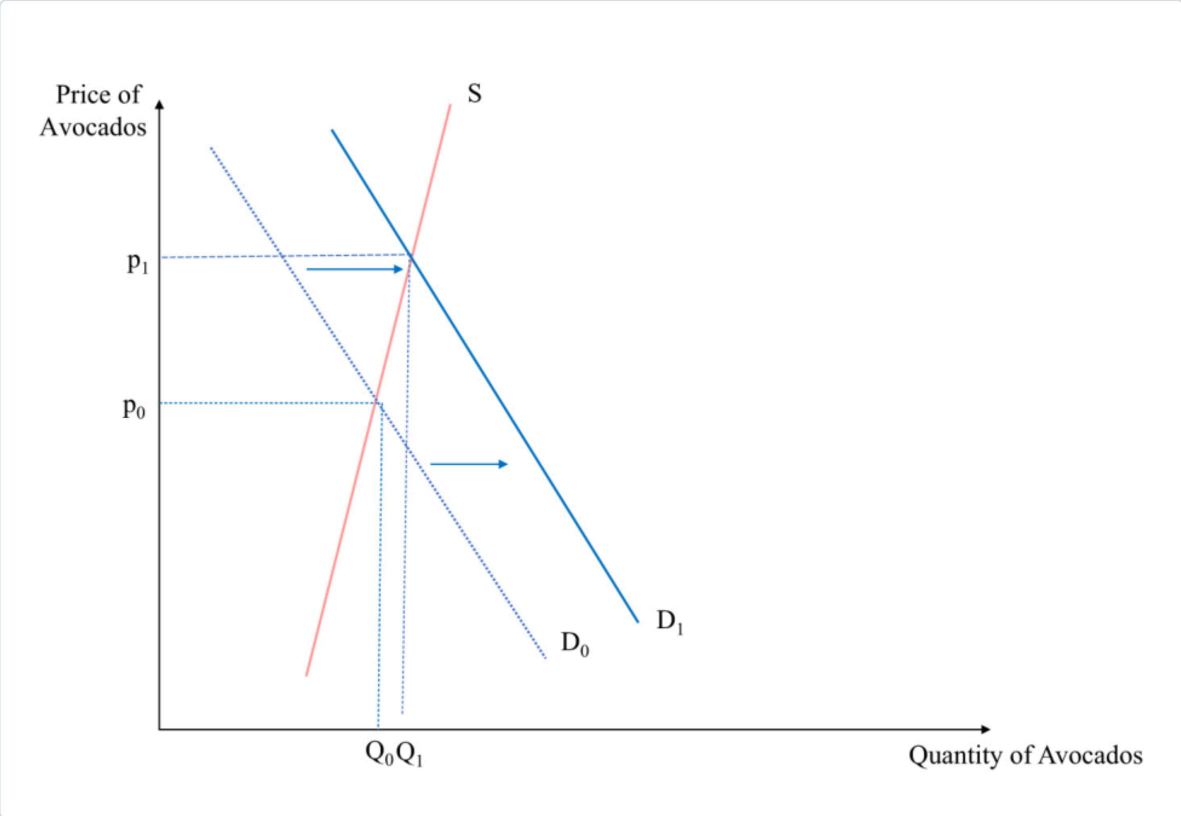
Core: to analyse any shock, **don't jump to price/quantity.** First figure out **which curve moves and which way**; the effect on p* and q* follows.

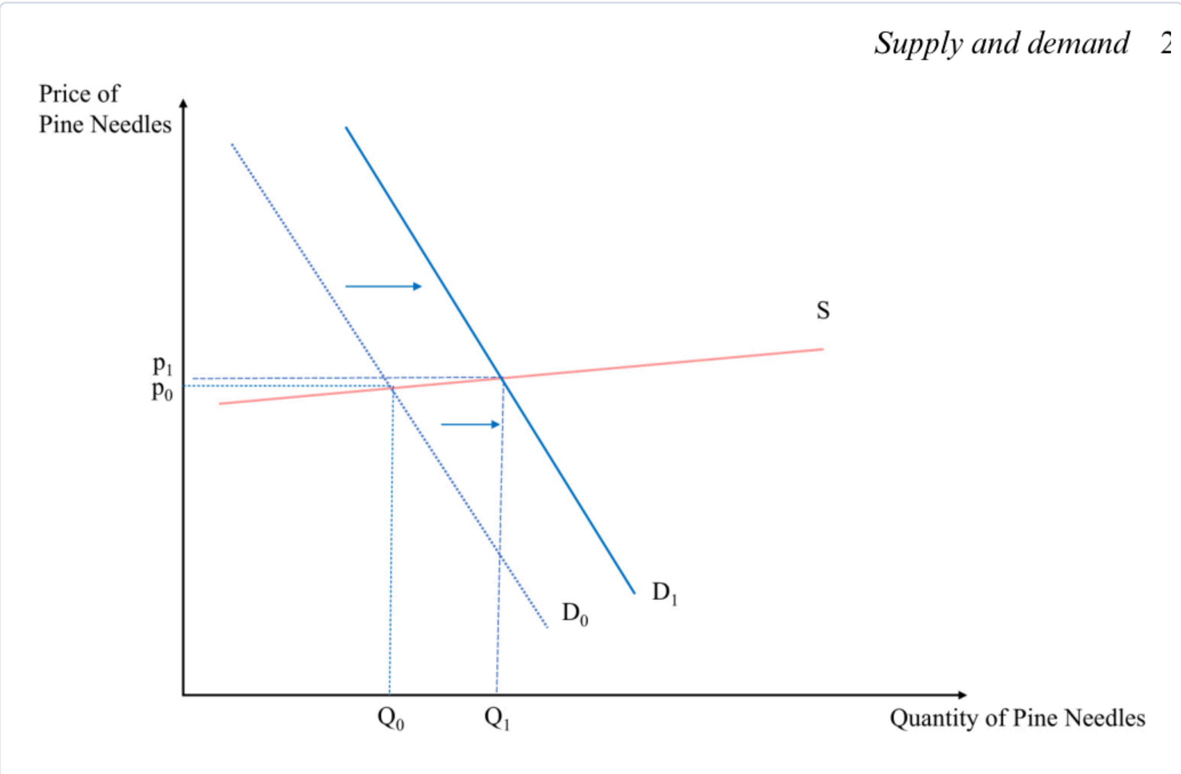
- **The method (Butera/Marshall — one change at a time, *ceteris paribus*):** ① **Demand or supply?** (falling incomes/other-good prices → demand; input costs/disruptions → supply). ② **Which direction** does it move *at a given price*? ③ *Then* read the new crossing. ④ **Reminder:** a *shift* is not a movement along — so a demand shift giving **both** higher price **and** higher quantity does **not** break the law of demand.



How to read it (a demand shock). Demand shifts **out** ($D_0 \to D_1$); supply is unchanged, so producers **move along** supply to the new cross — **both p and q rise**. (Add a simultaneous supply **fall** and price rises *unambiguously*, but the **quantity effect becomes ambiguous** — it depends which shift is bigger.)

- **★ Slope decides the *size* of the effect.** A **demand** shock's impact depends on the **slope of the *supply* curve**:





How to read them. Same outward demand shift, opposite results. **Steep supply** (avocados — trees take years, stock is fixed short-run): price jumps, quantity barely moves. **Flat supply** (pine needles — anyone can gather more): quantity jumps, price barely moves. *The flatter the other curve, the more a shock shows up as **quantity**; the steeper, the more it shows up as **price**.*

Price ceilings & floors — when the government fixes the price (deck)

Core: a **binding price control** stops the market clearing, and you find the resulting gap with the **exact excess-demand/supply calculation above** — just set p to the controlled price.

- **Price ceiling** = a legal **maximum** (e.g. **rent control**). To bind it must be **below p***, so **Q_D > Q_S → a shortage** (queues, waiting lists, black-market resale). *Compute:* plug the capped p into both curves; shortage = Q_D − Q_S.
- **Price floor** = a legal **minimum** (e.g. the **minimum wage** in the labour market). To bind it must be **above p***, so **Q_S > Q_D → a surplus** (unsold goods; in the labour market, **unemployment** — more people want to work than firms will hire). *Compute:* surplus = Q_S − Q_D.
- **The trade-offs (positive, not normative):** *ceilings* — **pro:** keep essentials affordable for the less-well-off; **con:** shortages, quality decline, and markets "find a way around" (rent control can even be **regressive**; Venezuela's caps produced empty shelves). *floors* — **pro:** a living wage; **con:** firms substitute **capital for labour** (self-checkout at Føtex), cut hours, or make jobs temporary. "*Good economics generates **positive** statements — evidence on costs and benefits — to inform the **normative** political choice.*"

Is the model realistic? — perfect competition (deck)

Core: the supply-demand model assumes **perfect competition** — "*every model is wrong, but useful.*"

- **Assumptions:** ① every agent is a **price-taker**; ② goods are **homogeneous** (perfect substitutes — one farm's wheat = another's; Shell petrol = Exxon petrol); ③ **perfect information** about prices/quality; ④ **low transaction costs**; (and many buyers & sellers). **Break these** and you need refinements — **monopoly, market failures** (later chapters).

Cases & examples

Each: *the example → the concept it teaches.*

- **Strawberries (the running example)** → the whole toolkit: `Q_D = 10 − p`, `Q_S = −2 + p`, equilibrium **(6, 4)**, and the €4/€8 shortage/surplus.
- **Diamond–water paradox** → price is set by **supply *and* demand** (Marshall's scissors), not usefulness — water's cheap because it's abundant.
- **Palladium** (tighter emission rules → catalytic-converter demand ↑; S. African mine disruptions → supply ↓) → **comparative statics** with one and two curves moving.
- **2008 world rice spike & Haiti riots** (India's export ban cut supply; panic stockpiling shifted demand out) → a real **double shift**; prices later fell as stockpiles released + the financial crisis hit demand.
- **Instant noodles** → **inferior good**; **raspberries** → **substitute**; **cereal & milk** → **complements**.
- **Giffen goods** (Irish-famine potatoes; poor Chinese rice households) → the rare **upward-sloping demand** exception.
- **COVID toilet-paper stockpiling** → an **expectations** shift of demand.
- **Avocados vs Swedish pine-needle tea** → **steep vs flat supply** sizing a demand shock.
- **Rent control / Venezuela** → **price ceiling → shortage**; **minimum wage** → **price floor → surplus (unemployment)**.

Formula sheet — quick reference

Object	Formula	Note
Demand function	$Q\_D = D(p, p_s, p\_c, Y, \tau)$	many variables; curve fixes all but p
Linear demand / inverse	$Q\_D = a - b \cdot p$ / $p = (a - Q\_D)/b$	a = intercept (shifters), b = slope
Supply function / inverse	$Q\_S = -2 + p$ / $p = 2 + Q\_S$	negative intercept = min price to supply
Aggregate (market)	$Q(p) = \Sigma D_i(p)$; $\Sigma S_i(p)$	**horizontal** sum (add quantities)
Equilibrium	**set Q_D = Q_S → p*, then q***	the market-clearing point
Excess demand (shortage)	$Q\_D - Q\_S$ at $p < p^*$	→ upward price pressure; price ceiling
Excess supply (surplus)	$Q\_S - Q\_D$ at $p > p^*$	→ downward pressure; price floor

Exam pointers

- **Solve a market from scratch:** given a linear demand and supply, **set them equal**, get **p***, back-substitute for **q*** — the strawberry case ($p^*=6$, $q^*=4$) is the model answer.
- **Compute a shortage/surplus:** plug a below- or above-equilibrium price into *both* curves and take the gap (**Q_D − Q_S** or **Q_S − Q_D**) — and know a **ceiling→shortage**, **floor→surplus**.
- **Diagnose movement vs shift**, and for a shock say **which curve moves, which way**, and the effect on p* & q* (incl. the **ambiguous-quantity** double-shift and the **slope** point).
- Explain **why demand slopes down** (scarcity + diminishing marginal utility) and **Marshall's scissors** / the diamond–water paradox.
- Name the **perfect-competition** assumptions and why the model is still "useful though wrong."

★ Fun facts & memorable details (Lecture 2)

> Sticky bits from Supply & Demand.

- **Marshall's scissors:** asking whether demand or supply sets the price is like asking **which blade of the scissors** does the cutting — both do.
- The **diamond–water paradox** is only a paradox if you forget supply: water is life-or-death yet cheap **because it's abundant**.
- **Giffen goods** really exist: when their staple got dearer, poor Chinese rice households bought **more** rice, not less (Jensen & Miller 2008) — they could no longer afford anything else.
- A negative supply intercept (**$Q\_S = -2 + p$**) isn't a mistake — it just means **price must top €2 before anyone supplies a drop.**
- **Vernon Smith** won a Nobel (2002) for showing that real people in a lab **converge on the equilibrium price** startlingly fast — the "Walrasian auctioneer" made flesh.

- Rent control's dirty secret: it can be **regressive** and spawn black-market "key money" — a "tale of good intentions" (cf. Venezuela's empty shelves).
- Minimum wage as a price floor nudges firms toward **capital over labour** — the **self-checkout at Føtex** is a supply-and-demand diagram in the wild.